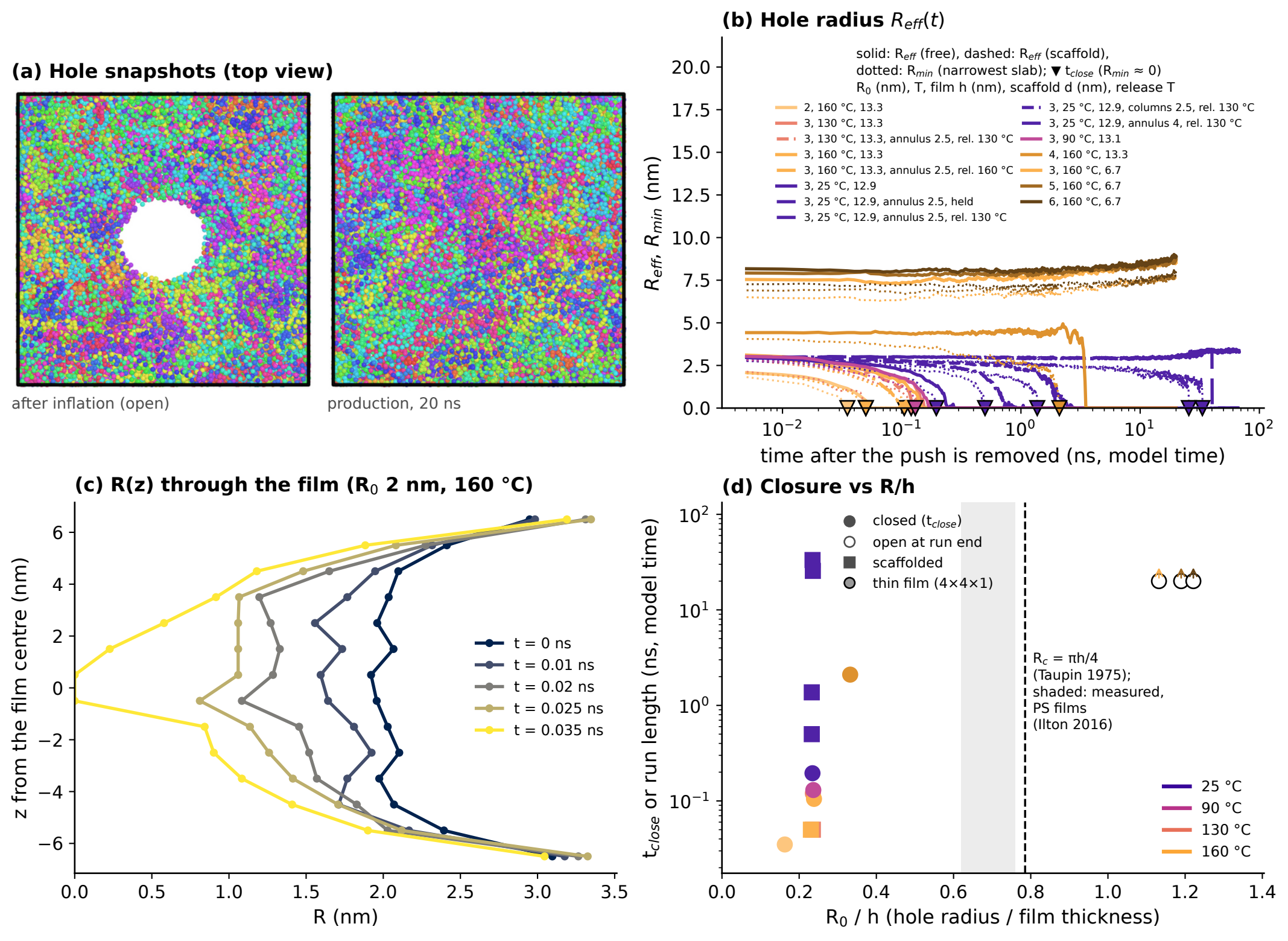

(a) Hole snapshots (top view)
after inflation (open)
production, 20 ns
(b) Hole radius $R_{eff}(t)$
solid: R_{eff} (free), dashed: R_{eff} (scaffold),
dotted: R_{min} (narrowest slab); ▼ t_{close} (R_{min} ≈ 0)
R_0 (nm), T, film h (nm), scaffold d (nm), release T
2, 160 °C, 13.3
3, 130 °C, 13.3
3, 130 °C, 13.3, annulus 2.5, rel. 130 °C
3, 160 °C, 13.3
3, 160 °C, 13.3, annulus 2.5, rel. 160 °C
3, 25 °C, 12.9
3, 25 °C, 12.9, annulus 2.5, held
3, 25 °C, 12.9, annulus 2.5, rel. 130 °C
3, 25 °C, 12.9, columns 2.5, rel. 130 °C
3, 25 °C, 12.9, annulus 4, rel. 130 °C
3, 90 °C, 13.1
4, 160 °C, 13.3
3, 160 °C, 6.7
5, 160 °C, 6.7
6, 160 °C, 6.7
R_{eff}, R_{min} (nm)
time after the push is removed (ns, model time)
(c) R(z) through the film (R_0 2 nm, 160 °C)
z from the film centre (nm)
R (nm)
t = 0 ns
t = 0.01 ns
t = 0.02 ns
t = 0.025 ns
t = 0.035 ns
(d) Closure vs R/h
t_{close} or run length (ns, model time)
R_0 / h (hole radius / film thickness)
closed (t_{close})
open at run end
scaffolded
thin film (4×4×1)
$R_c = \pi h/4$
(Taupin 1975);
shaded: measured,
PS films
(Ilton 2016)
25 °C
90 °C
130 °C
160 °C